Neural Networks and Binary Data

Santiago Miranda

PhD Application

March 12, 2026

Areas of experience

Deep Learning

Scientific Computing

Areas of experience

Deep Learning

- Molecular inverse design with VAEs

Scientific Computing

Areas of experience

Deep Learning

- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

Areas of experience

Deep Learning

- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

- Binary parsers for spectroscopy data

Areas of experience

Deep Learning

- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

- Binary parsers for spectroscopy data
- Mathematical optimisation algorithms

Areas of experience

Deep Learning

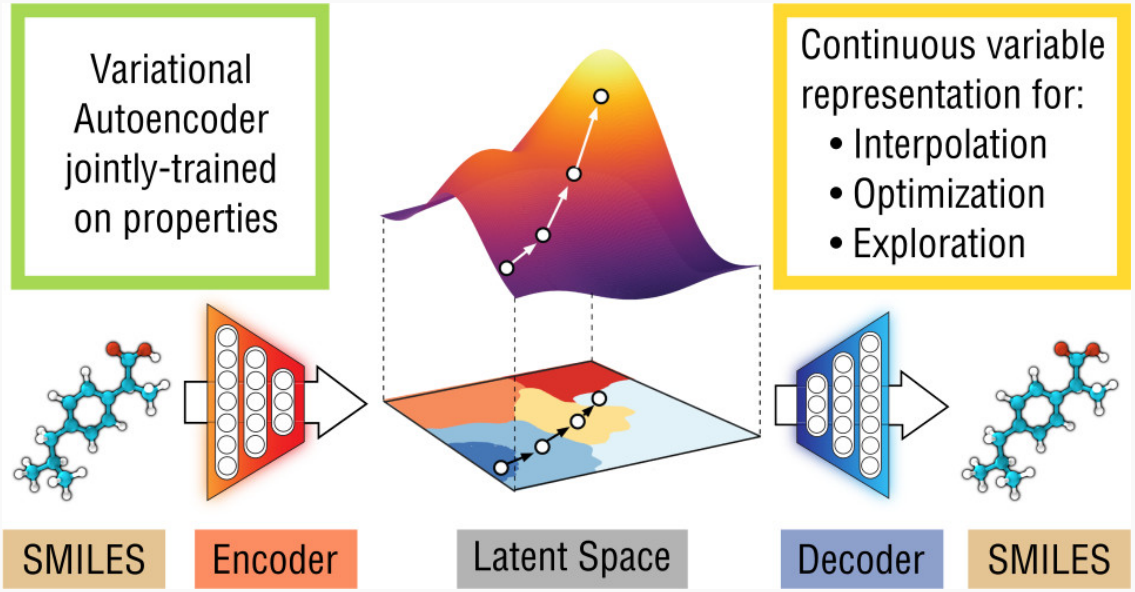
- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

- Binary parsers for spectroscopy data
- Mathematical optimisation algorithms

Generative AI - Inspiration

Figure: VAE: Aspuru-Guzik et. al.



- Learns a continuous vector representation;
- Can the latent space be interpretable? How to enhance valid-decoding rate?

Generative AI - Project Results

First 2–3 months:[1]

Completed:

Review	literature of deep learning models → selected VAE ,
Implement	simplified model (Keras 3),
Deploy	quantised ONNX model in browser,
Package	document, test, notebooks.

[1] Work carried out at the University of Glasgow.

Generative AI - Project Results

First 2–3 months:[1]

Completed:

- **Review** literature of deep learning models → selected VAE ,
- **Implement** simplified model (Keras 3),
- **Deploy** quantised ONNX model in browser,
- **Package** document, test, notebooks.

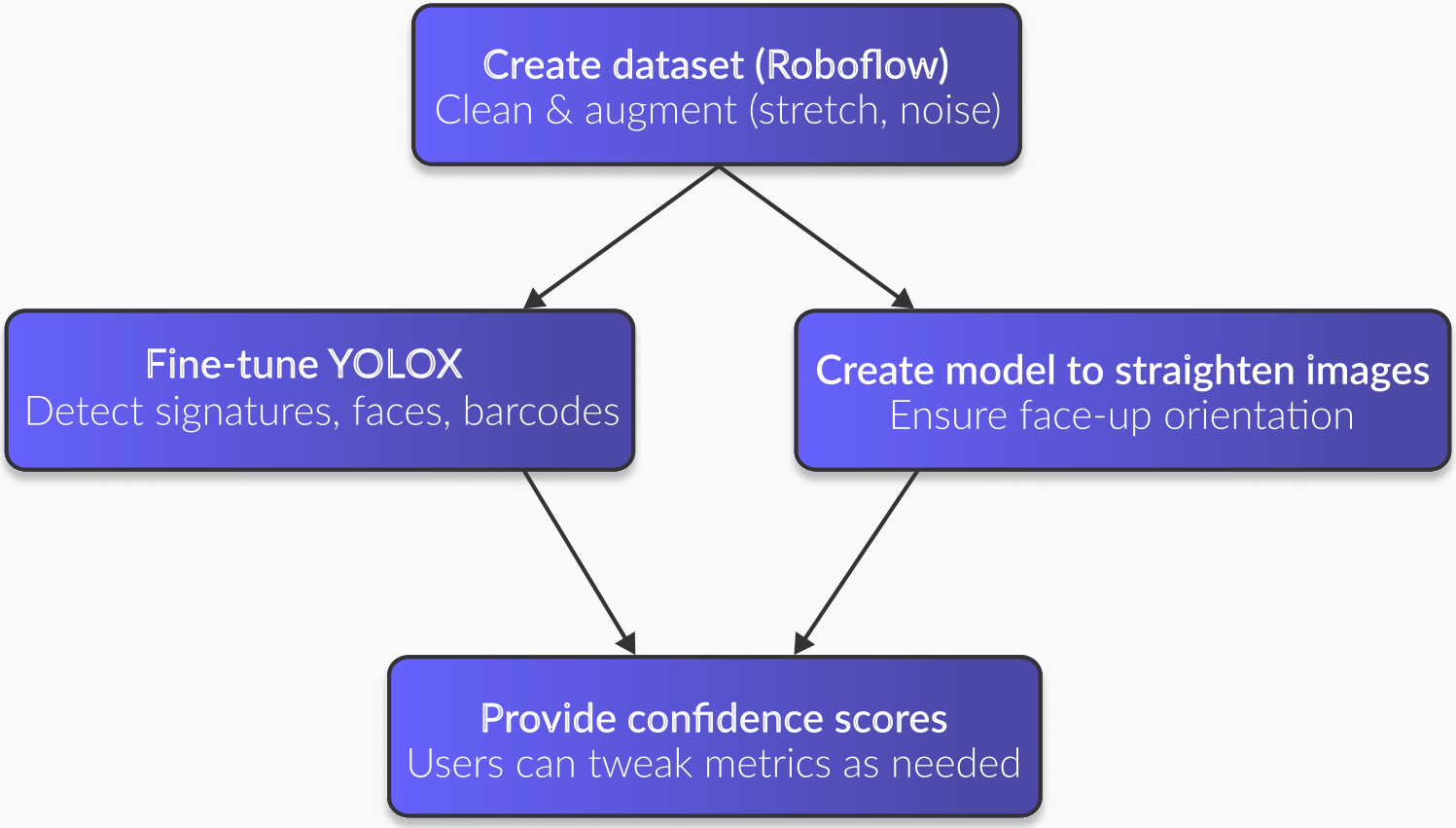
Planned:

- **Optimise** UV HOMO-LUMO gap,
- **Select** candidates for robotic synthesis.

[1] Work carried out at the University of Glasgow.

Object Detection - Overview

Figure: Project's Development Stages



Megvii-BaseDetection/yolox

Object Detection - Results

- Pseudonymisation of IDs at Zakodium (demo site)
- Frameworks: Tensorflow.js, Pytorch

Object Detection - Results

- Pseudonymisation of IDs at Zakodium (demo site)
- Frameworks: Tensorflow.js, Pytorch

(a) Face and signature detection

Object Detection - Results

- Pseudonymisation of IDs at Zakodium (demo site)
- Frameworks: Tensorflow.js, Pytorch

(a) Face and signature detection

(b) Face, barcode, MRZ detection

Object Detection - Results

- Pseudonymisation of IDs at Zakodium (demo site)
- Frameworks: Tensorflow.js, Pytorch

(a) Face and signature detection

(b) Face, barcode, MRZ detection

Figure: © European Union, 1998–2026.

Binary Data Manipulation - iobuffer

`iobuffer` is a helper to write binary parsers.

```
import { IOBuffer } from 'iobuffer';

const io = new IOBuffer() // offset 0
io.writeChars('Hello world') // offset now 11
  .back(2) // Back to 9
  .skip(1) // Forward to 10
  .rewind() // back to 0
  .readArray(11, "uint8") // read 11 bytes at once
```

Contributions: text decoder, readArrays of typed-data, host-endianness detection.

Other Projects

Github Project	npm ↓	Description
iobuffer	downloads 19M/month	Binary data manipulation library
regression-simple-linear	downloads 589k/month	Simple linear regression
regression-polynomial	downloads 511k/month	Polynomial regression
fcnnls	downloads 3.2k/month	Fast combinatorial non-negative least squares
spc-parser	downloads 1.3k/month	GRAMS .spc parser
wdf-parser	downloads 804/month	WDF Raman file parser
IDs Project	–	Pseudonymisation of IDs.

Table: Github hosted projects, excepting last (IDs) project.

Thanks!